

STACK_AND_SKILLS.md

Mandatory Technology Stack

Language

Primary Language:

Go

Reason:

- Concurrency
 - Performance
 - Simplicity
 - Cloud Native Ecosystem
-

Backend Libraries

CLI:

- Cobra
- Viper

Metrics:

- Prometheus Go Client

Configuration:

- YAML v3

Logging:

- Zap

Testing:

- Testify

HTTP:

- net/http

Terminal UI:

- BubbleTea

- LipGloss
-

Infrastructure

Containerization:

- Docker

Registry:

- GHCR

Orchestration:

- Kubernetes

Packaging:

- Helm

GitOps:

- ArgoCD

Monitoring:

- Prometheus
- Grafana

IaC:

- Terraform

Cloud:

- AWS
-

Engineering Standards

Code Quality

Requirements:

- golangci-lint
- unit tests
- integration tests

Coverage:

Minimum 80%

Security

Requirements:

- Trivy scans
 - SBOM generation using Syft
 - No secrets in repository
-

Observability

Requirements:

- Structured logs
 - Prometheus metrics
 - Grafana dashboards
-

Kubernetes Standards

Requirements:

- Health Checks
 - Readiness Probes
 - Liveness Probes
 - Resource Requests
 - Resource Limits
-

Terraform Standards

Requirements:

- Remote State
- State Locking
- Modules
- Separate Environments

Environments:

- dev
 - staging
 - production
-

Required Skills For Contributors

Go Concurrency

- Goroutines
- Channels
- Contexts

Networking

- HTTP
- TLS
- DNS

Cloud Native

- Containers
- Kubernetes
- Helm
- ArgoCD

Observability

- Metrics
- Logging
- Tracing

Infrastructure

- Terraform
- AWS

Performance Engineering

- Benchmarking
- Profiling
- Load Testing

Definition of Done

A feature is complete only if:

✓ Unit Tests Pass

✓ Integration Tests Pass

✓ Lint Passes

✓ Docker Build Passes

- ✓ Trivy Passes
- ✓ Prometheus Metrics Exposed
- ✓ Documentation Updated
- ✓ Helm Chart Updated
- ✓ CI/CD Pipeline Green